EE502 - Linear Systems Theory II **Spring 2023**

Lecture 14

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14.1 State Observer

Generally the full state measurement, $x(t)$ or $x[k]$, of a system is not accessible and (observers, estimators, filters) have to be used to extract this information. The output, $y(t)$ or $y[k]$, represents the actual measurements which is a function of the input and the output. We will mainly focus on DT systems in this lecture; however, all of the derivations and concepts are either same with the CT case or can be easily adopted with minor modifications. Let

$$\begin{aligned} x[k+1] &= Ax[k] + Bu[k] \\ y[k] &= Cx[k] + Du[k] \end{aligned}$$

A Luenberger observer is built using a "simulated" model of the system and the errors caused by the mismatched initial conditions $x_0 \neq \hat{x}_0$ (or other types of perturbations) are reduced by introducing output error feedback.

Let's assume that the state vector of the simulated system is $\hat{x}[k]$, then the state space equation of this synthetic system takes the form

$$\begin{aligned} \hat{x}[k+1] &= A\hat{x}[k] + Bu[k] \\ \hat{y}[k] &= C\hat{x}[k] + Du[k] \end{aligned}$$

Note that since $u[k]$ is the input that is supplied by the controller, we assume that it is known a priori. If $x[0] = \hat{x}[0]$ and when there is no model mismatch or uncertainty in the system, then we expect that $x[k] = \hat{x}[k]$ and $y[k] = \hat{y}[k]$ for all k. When $x[0] \neq \hat{x}[0]$, then we should observe a difference between the measured and predicted output $y[k] \neq \hat{y}[k]$. The core idea in a Luenberger observer is feeding the error in the output prediction $y[k] - \hat{y}[k]$ to the simulated system via a linear feedback gain.

$$\begin{aligned} \hat{x}[k+1] &= A\hat{x}[k] + Bu[k] + L\left(y[k] - \hat{y}[k]\right) \\ \hat{y}[k] &= C\hat{x}[k] + Du[k] \end{aligned}$$

In order to understand how a Luenberger observer works and to choose a proper observer gain L, we define an error signal $e[k] = x[k] - \hat{x}[k]$. The dynamics w.r.t $e[k]$ can be derived as

$$\begin{aligned} e[k+1] &= x[k+1] - \hat{x}[k+1] \\ &= \left(Ax[k] + Bu[k]\right) - \left(A\hat{x}[k] + Bu[k] + L\left(y[k] - \hat{y}[k]\right)\right) \\ e[k+1] &= \left(A - LC\right) e[k] \end{aligned}$$

where $e[0] = x[0] - \hat{x}[0]$ denotes the error in the initial condition.

If the matrix $(A - LC)$ is stable then the errors in initial condition will diminish eventually. Moreover, in order to have a good observer/estimator performance the observer convergence should be sufficiently fast.

Theorem: (Observer Eigenvalue Placement) Given (A, C), $\exists L$ s.t.

$$\det\left[\lambda I - (A - LC)\right] = \lambda^n + a^*_{n-1}\lambda^{n-1} + \cdots + a^*_1\lambda + a^*_0$$
$$\forall \mathcal{A} = \{a^*_0,\, a^*_1 \cdots a^*_{n-1}\},\, a^*_i \in \mathbb{R}$$

if and only if (A, C) is observable.

Proof of necessity: Let's assume that (A, C) not observable and $\exists(\lambda_u, v_u)$ pair such that $Av_u = v_u\lambda_u$ and $Cv_u = 0$. Now check whether v_u is a right eigenvector of $(A - LC)$

$$(A - LC)v_u = Av_u - LCv_U = Av_u = \lambda_u v_U$$
$$Cv_u = 0$$

Here not only we showed that λ_u can not be moved hence it is not possible to locate the observer poles to arbitrary locations, we also showed that state-observer rule does not affect the observability.

Proof of sufficiency: Note that if (A, C) pair is observable then (A^T, C^T) pair is reachable. Then we know that $\exists L^T$ s.t.

$$\det\left[\lambda I - (A^T - C^T L^T)\right] = \lambda^n + a^*_{n-1}\lambda^{n-1} + \cdots + a^*_1\lambda + a^*_0$$
$$\forall \mathcal{A} = \{a^*_0,\, a^*_1 \cdots a^*_{n-1}\},\, a^*_i \in \mathbb{R}$$

and obviously

$$\det\left[\lambda I - (A^T - C^T L^T)\right] = \det\left[\lambda I - (A - LC)\right]$$

which completes the proof.

14.2 Detectability

The dual of stabilizability in the observability domain is called detectability.

Theorem The following statements are equivalent

- (A, C) is detectable $\iff$
- If unobservable modes of (A, C) are asymptotically stable $\iff$
- $\forall(\lambda_i, v_i)$ s.t. $Av_i = \lambda_i v_i \iff$ and $\mathrm{Re}\{\lambda_i\} \geq 0$ (or $|\lambda_i| \geq 0$), $v_i \notin \mathcal{N}(C) \iff$
- rank $\left[\dfrac{A - \lambda I}{C}\right] = n$, $\forall \lambda \in \mathbb{C}$ s.t. $\mathrm{Re}\{\lambda_i\} \geq 0$ for CT systems (or $\forall \lambda \in \mathbb{C}$ s.t. $|\lambda| \geq 1$ for DT systems)
- $\exists L$ such that $(A - LC)$ is asymptotically stable

14.3 Model Based Controllers: Closing the loop between the observer and state-feedback control low

In the state-feedback control policy, we defined the control law as

$$u[k] = \alpha r[k] - Kx[k]$$

However, as mentioned in the Observer section, in general, we don't have direct access to all states of the system. In this case, we learned how to design an Observer/Estimator that predicts the states of the system. In this respect, it is natural to assume that in a closed-loop system, the control policy that defines the input should depend on the estimated states.

$$u[k] = \alpha r[k] - K\hat{x}[k]$$

However, the critical question is how this coupling affects the closed-loop behavior, and an even deeper question is can we even use such a policy. The advantage of LTI systems is that state-feedback gain and observer gain can be separately designed, and we guarantee a stable closed-loop performance. In this section, we will analyze the coupled system. Equations of motion for the closed-loop observer & state-feedback based control system is given below

$$\begin{aligned}
x[k+1] &= Ax[k] + Bu[k] \\
y[k] &= Cx[k] + Du[k] \\
\hat{x}[k+1] &= A\hat{x}[k] + Bu[k] + L\left(y[k] - \hat{y}[k]\right) \\
\hat{y}[k] &= C\hat{x}[k] + Du[k] \\
u[k] &= -K\hat{x}[k]
\end{aligned}$$

If we eliminate $u[k]$ and $\hat{y}[k]$ we obtain following dynamical representation

$$\begin{aligned}
x[k+1] &= Ax[k] - BK\hat{x}[k] + B\alpha r[k] \\
\hat{x}[k+1] &= A\hat{x}[k] - BK\hat{x}[k] + LC\left(x[k] - \hat{x}[k]\right) + B\alpha r[k] \\
y[k] &= Cx[k] - DK\hat{x}[k] + D\alpha r[k]
\end{aligned}$$

Now let's replace $\hat{x}[k]$ with $e[k] = x[k] - \hat{x}[k]$

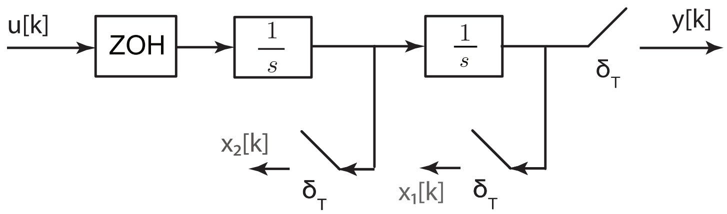
$$\begin{aligned}
x[k+1] &= (A - BK)x[k] + BKe[k] + B\alpha r[k] \\
e[k+1] &= (A - LC)e[k] \\
y[k] &= (C - DK)x[k] + DKe[k] + D\alpha r[k]
\end{aligned}$$

Now let's define a state for the whole system, $z[k] = \begin{bmatrix} x[k] \\ e[k] \end{bmatrix}$ then the state-space representation is given by

$$\begin{aligned}
z[k+1] &= \begin{bmatrix} (A - BK) & BK \\ 0_{n\times n} & (A - LC) \end{bmatrix} z[k] + \begin{bmatrix} \alpha B \\ 0 \end{bmatrix} r[k] \\
y[k] &= \begin{bmatrix} (C - DK) & DK \end{bmatrix} z[k] + [\alpha D] r[k]
\end{aligned}$$

The system matrix is in block diagonal form, and the eigenvalues of this new system, the matrix is found by taking the union of eigenvalues of $(A - BK)$ and eigenvalues of $(A - LC)$. Thus a separate pole-placement can be performed for the state-feedback controller and the observer.

Ex 14.1 *Consider the following open-loop digital control systems*



1. Find the discrete-time state-space representation between the discrete time input signal, $u[k]$ and the discrete time output signal, $u[k]$, given that state definition is $x[k] = \begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix}$. Note that sampling time is generic for this problem T.

   Solution: The state update equation of the continuous-time plant has the following form

$$\dot{x} = A_{ct}x + B_{ct}u \ , \ x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$
$$A_{ct} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \ , \ B_{ct} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

   Since the ADC block is a simple ZOH operator we can find the system matrix of the discritized system using the state-transitions matrix of the CT-Plant

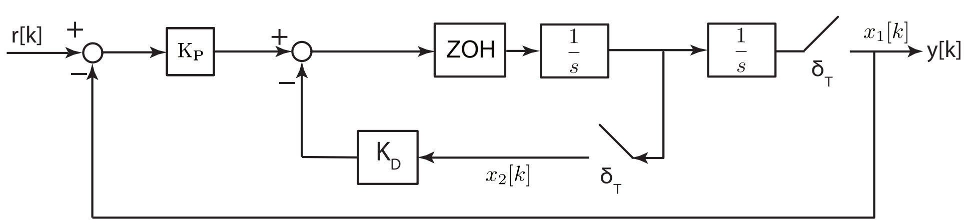
$$\begin{aligned} A_{dt} &= e^{A_{ct}T} = I + A_{ct}T + 0 + \cdots \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & T \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \end{aligned}$$

   Since A_{ct} is not invsertable, we need to derive B_{dt} using the following relation

$$B_{dt} = \int_0^T e^{A_{ct}\tau} B_{ct} d\tau = \int_0^T \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau = \int_0^T \begin{bmatrix} \tau \\ 1 \end{bmatrix} d\tau = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}$$

   Output equation is simply equal to $y[k] = x_1[k]$, and thus $C_{dt} = C_{ct} = C = \begin{bmatrix} 1 & 0 \end{bmatrix}$ and $D_{dt} = D_{ct} = CD = \begin{bmatrix} 0 \end{bmatrix}$.

2. Show that following closed-loop PD type controller is indeed a form of state-feedback rule



   Solution: Let's explicitly write $u[k]$

$$\begin{aligned} u[k] &= K_P(r[k] - x_1[k]) + K_D x_2[k] = K_P r[k] - \begin{bmatrix} K_P & K_D \end{bmatrix} x[k] \\ K &= \begin{bmatrix} K_P & K_D \end{bmatrix} \ , \ \alpha = K_P \end{aligned}$$

3. Show that we can always find a (K_P, K_D) pair such that closed-loop poles are located at arbitrary desired locations.

   Solution: We need to simply show that (A_{dt}, B_{dt}) pair is reachable

$$\mathbf{R} = \left[\begin{array}{c|c} A_{dt}B_{dt} & B_{dt} \end{array}\right] = \left[\begin{array}{c|c} 3T^2/2 & T^2/2 \\ T & T \end{array}\right] \Rightarrow \text{rank}[\mathbf{R}] = 2$$

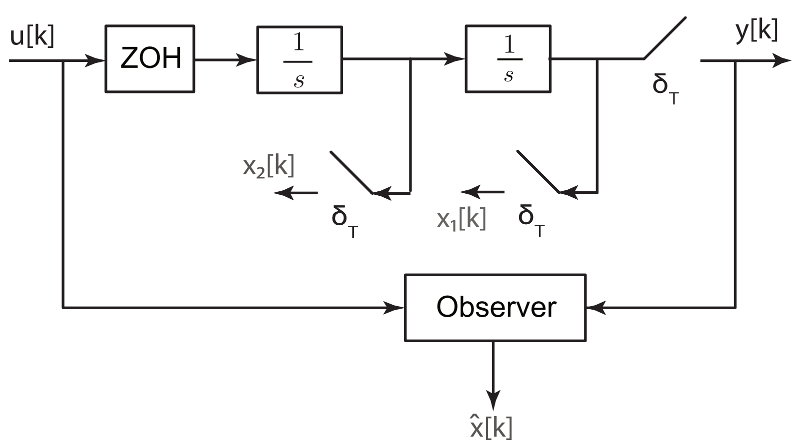
   thus (A_{dt}, B_{dt}) reachable and it implies that we can always find a (K_P, K_D) pair such that closed-loop poles are located at arbitrary desired locations.

4. Find (K_P, K_D) such that closed-loop systems shows dead-beat behavior

   Solution: In such a case, we need to place the closed-loop eigenvalues at the origin

$$\begin{aligned}
A_{cl} = A_{dt} - B_{dt}K &= \begin{bmatrix}1 & T\\0 & 1\end{bmatrix} - \begin{bmatrix}T^2/2\\T\end{bmatrix}\begin{bmatrix}K_P & K_D\end{bmatrix} = \begin{bmatrix}1 & T\\0 & 1\end{bmatrix} - \begin{bmatrix}K_PT^2/2 & K_DT^2/2\\K_PT & K_DT\end{bmatrix}\\
&= \begin{bmatrix}1-K_PT^2/2 & T-K_DT^2/2\\-K_PT & 1-K_DT\end{bmatrix}\\
\lambda I - A_{cl} &= \begin{bmatrix}\lambda-1+K_PT^2/2 & -T+K_DT^2/2\\K_PT & \lambda-1+K_DT\end{bmatrix}\\
\det(\lambda I - A_{cl}) &= \lambda^2 + \left(K_PT^2/2 + K_DT - 2\right)\lambda + \left(K_PT^2/2 - K_DT + 1\right)\\
\begin{bmatrix}2\\-1\end{bmatrix} &= \begin{bmatrix}T^2/2 & T\\T^2/2 & -T\end{bmatrix}\begin{bmatrix}K_P\\K_D\end{bmatrix} \Rightarrow \begin{bmatrix}K_P\\K_D\end{bmatrix} = \frac{1}{T}\begin{bmatrix}T/2 & 1\\T/2 & -1\end{bmatrix}^{-1}\begin{bmatrix}2\\-1\end{bmatrix} = \frac{1}{T}\begin{bmatrix}1/T & 1/T\\1/2 & -1/2\end{bmatrix}\begin{bmatrix}2\\-1\end{bmatrix}\\
\begin{bmatrix}K_P\\K_D\end{bmatrix} &= \begin{bmatrix}1/T^2\\3/(2T)\end{bmatrix} \Rightarrow K = \begin{bmatrix}1/T^2 & 3/(2T)\end{bmatrix}\\
A_{cl} &= \begin{bmatrix}1/2 & T/4\\-1/T & -1/2\end{bmatrix}
\end{aligned}$$

5. Now design observer as shown in the figure below such that the observer states converge the actual state in a deadbeat fashion.



   Solution: Let's first check weather the system is observable or not

$$\mathbf{O} = \left[\begin{array}{c} C \\ \hline CA_{dt} \end{array}\right] = \left[\begin{array}{cc} 1 & 0 \\ \hline 1 & T \end{array}\right], \ \text{rank}(\mathbf{O}) = 2 \Rightarrow \text{observable}$$

Now let's design the observer. We know that observer's state-space model has the following form

$$\hat{x}[k+1] = A_{dt}\hat{x}[k] + B_{dt}u[k] + L\left(y[k] - C\hat{x}[k]\right)$$

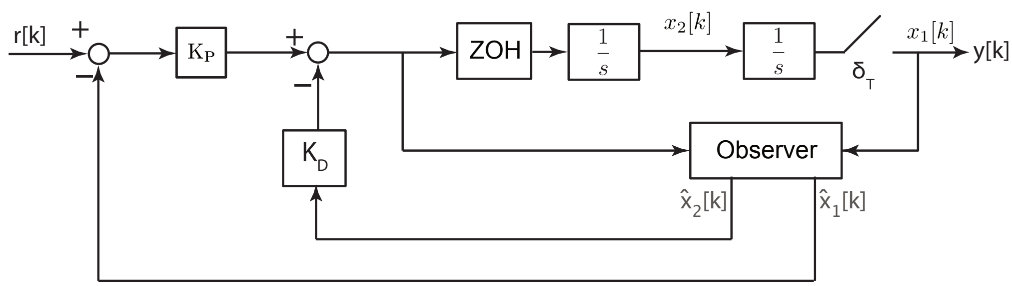
and the error dynamics between the actual and predicted states has the following autonomous structure

$$e[k+1] = (A_{dt} - LC)e[k]$$

In order to obtain dead-beat behavior in error dynamics

$$\begin{aligned}
\det(\lambda I - A_o) &= \lambda^2 \ , \ A_o = (A_{dt} - LC) \\
(\lambda I - A_o) &= \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} \lambda - 1 + l_1 & -T \\ l_2 & \lambda - 1 \end{bmatrix} \\
\det(\lambda I - A_o) &= \lambda^2 + (l_1 - 2)\lambda + (-l_1 + Tl_2 + 1) \\
L &= \begin{bmatrix} l_1 \\ l_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1/T \end{bmatrix} \Rightarrow A_o = \begin{bmatrix} -1 & T \\ -1/T & 1 \end{bmatrix}
\end{aligned}$$

6. Now close the loop using the observers estimated states as shown i the figure below. Fine the state-space representation of the whole closed-loop system and comment on the transient behavior of the dynamics



Solution: In the lecture notes for a closed-loop observer and state-feedback structure, we already derived the full state-space form

$$z[k+1] = \begin{bmatrix} (A_{dt} - B_{dt}K) & B_{dt}K \\ 0_{n\times n} & (A_{dt} - LC) \end{bmatrix} z[k] + \begin{bmatrix} \alpha B_{dt} \\ 0 \end{bmatrix} r[k] \ , \ y[k] = \begin{bmatrix} C & 0 \end{bmatrix} z[k]$$

$$z[k+1] = \left[\begin{array}{cc|cc} 1/2 & T/4 & 1/2 & 3T/4 \\ -1/T & -1/2 & 1/T & 3/2 \\ \hline 0 & 0 & -1 & T \\ 0 & 0 & -1/T & 1 \end{array}\right] z[k] + \left[\begin{array}{c} 1/2 \\ 1/T \\ \hline 0 \\ 0 \end{array}\right] r[k]$$

$$y[k] = y[k] = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 0 \end{array}\right] z[k]$$